

An Exact Computer-Assisted Perfectness Census for Integral Circulant Graphs Through Order 32

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Abstract

For every integer $1 \leq n \leq 32$ and every subset D of the strict positive divisors of n , we determine whether the integral circulant graph $\text{ICG}_n(D)$ is perfect. There are 539 such pairs: 415 are perfect and 124 are imperfect. Imperfect examples occur only at orders 12, 20, 24, 28, and 30, and a complete divisor-mask table is given. The classification is exact and computer-assisted. Each imperfect record contains an induced odd hole in the graph or its complement. Each perfect record is certified by exhaustive induced-path searches on both sides; translation symmetry reduces a complete search to cycles containing vertex 0, and the Strong Perfect Graph Theorem then applies. Separate adjacency and witness replay and a distinct chordless-cycle implementation corroborate the census. No conclusion is drawn for $33 \leq n \leq 64$ or for unbounded n ; in particular, the broader classification problem remains unresolved.

1 Scope and context

Let

$$\mathcal{D}_n = \{d \in \mathbb{Z}_{>0} : d \mid n, d < n\}.$$

For $D \subseteq \mathcal{D}_n$, the integral circulant graph $\text{ICG}_n(D)$ has vertex set $\mathbb{Z}/n\mathbb{Z}$, with distinct x, y adjacent exactly when

$$\gcd(x - y, n) \in D.$$

This is the usual divisor-set description of integral circulant graphs [4, 1]. A graph is perfect when every induced subgraph has chromatic number equal to clique number. By the Strong Perfect Graph Theorem, it is perfect exactly when neither it nor its complement contains an induced odd cycle of length at least five [2].

The unitary subfamily $D = \{1\}$ has a known perfectness classification [3]. So's earlier work enumerated integral-circulant symbols below 100 for spectral purposes [4]; that is prior enumeration of the underlying inputs, but it is not a perfectness census with forbidden-cycle and positive certificates. Recent work again records the classification for arbitrary divisor sets as a broader direction [1]. The contribution here is deliberately finite: an exact, replayable classification of the complete $n \leq 32$ layer. A bounded source comparison found no equivalent perfectness table, but this negative search is not a claim of priority.

Write the elements of $\mathcal{D}_n$ increasingly as $d_0, \dots, d_{k-1}$. We encode D by

$$m(D) = \sum_{d_i \in D} 2^i.$$

Thus $0 \leq m(D) < 2^k$, and the pair $(n, m(D))$ determines D without ambiguity.

2 The census

Theorem 1 (Complete classification through order 32). *For $1 \leq n \leq 32$ and $D \subseteq \mathcal{D}_n$, the graph $\text{ICG}_n(D)$ is imperfect if and only if $(n, m(D))$ occurs in Table 1. Consequently, among all 539 pairs (n, D) there are exactly 415 perfect and 124 imperfect graphs. Imperfect graphs occur only for $n \in \{12, 20, 24, 28, 30\}$.*

Table 1: All imperfect divisor masks. Divisors are ordered increasingly, and every mask not listed is perfect.

n	$\mathcal{D}_n$	imperfect masks $m(D)$
12	[1, 2, 3, 4, 6]	6, 9, 12, 19, 22, 25
20	[1, 2, 4, 5, 10]	5, 10, 12, 19, 21, 26
24	[1, 2, 3, 4, 6, 8, 12]	6, 9, 12, 14, 19, 22, 24, 25, 27, 28, 29, 30, 33, 34, 35, 36, 38, 39, 41, 44, 48, 49, 51, 52, 53, 54, 57, 70, 73, 74, 75, 76, 78, 79, 83, 86, 88, 89, 91, 92, 93, 94, 97, 98, 99, 100, 102, 103, 105, 108, 113, 115, 118, 121
28	[1, 2, 4, 7, 14]	5, 10, 12, 19, 21, 26
30	[1, 2, 3, 5, 6, 10, 15]	6, 10, 14, 15, 17, 19, 21, 22, 24, 27, 28, 30, 33, 35, 36, 39, 41, 42, 44, 46, 49, 52, 53, 56, 57, 61, 66, 70, 71, 74, 75, 78, 81, 83, 85, 86, 88, 91, 92, 94, 97, 99, 100, 103, 105, 106, 108, 110, 112, 113, 117, 121

The table includes all boundary cases through its complement: $n = 1$, $D = \emptyset$, $D = \mathcal{D}_n$, disconnected graphs, complete graphs, and their complements were not removed. Indeed,

$$\overline{\text{ICG}_n(D)} = \text{ICG}_n(\mathcal{D}_n \setminus D), \quad (1)$$

because the gcd of the difference of two distinct residues is always a strict divisor of n . Complement pairing was used as a check, not to reduce the stated input domain.

3 Exact certification

We describe the decisive exhaustive routine, rather than relying on a software perfectness flag. Every adjacency matrix is constructed directly from the gcd definition using integer arithmetic. For an adjacency relation A on $\mathbb{Z}/n\mathbb{Z}$ and each odd target length $\ell \in \{5, 7, \dots\} \cap [5, n]$, the search starts with the path (0) and tries every neighbor as a second vertex. Thereafter it maintains a simple induced path. A candidate adjacent to the current endpoint is handled as follows:

- if it is adjacent to 0, it is accepted only when it closes a path of exactly ℓ vertices and has no edge to an earlier vertex other than the endpoint and 0;
- if it is not adjacent to 0, it is appended only when the new path has fewer than ℓ vertices and it has no edge to any earlier vertex other than the endpoint.

The routine is run first on the graph and, if no odd hole is found there, on its complement.

Lemma 2 (Completeness of the translation-reduced search). *For a circulant graph G on $\mathbb{Z}/n\mathbb{Z}$, the routine above returns an induced odd cycle if and only if G has an induced odd cycle of length at least five.*

Proof. The extension rule maintains an induced path. At acceptance, the final vertex has precisely the two required cycle neighbors and no chord, so every returned sequence is an induced cycle.

Conversely, suppose G has an induced cycle C of odd length $\ell \geq 5$. Translation by the negative of any vertex of C is an automorphism of a circulant graph, so some translate of C contains 0. Orient this translated cycle as $(0, v_1, \dots, v_{\ell-1})$. The search tries v_1 . Each v_i with $i < \ell - 1$ is adjacent to its predecessor, nonadjacent to 0, and nonadjacent to every earlier nonpredecessor, so the corresponding branch is not pruned. The last vertex is adjacent to $v_{\ell-2}$ and 0 and to no other earlier vertex, hence it is accepted at target length ℓ . Searching all possible odd target lengths proves completeness. $\square$

The complement of a circulant graph is circulant by (1), so the lemma applies on both sides. Therefore a null result from both searches proves that the graph is Berge, and the Strong Perfect Graph Theorem proves it perfect. A returned cycle on either side proves imperfection.

Proposition 3 (Exact run and certificates). *The deterministic exhaustive run covered all 539 pairs and produced the classification in Table 1. It visited 2,789,071 induced-path search states. The 124 imperfect records contain 105 graph-side and 19 complement-side witnesses; 86 witnesses have length 5, 36 have length 7, and 2 have length 9.*

Proof. For fixed n , the masks $0, \dots, 2^{|\mathcal{D}_n|} - 1$ are in bijection with all subsets of $\mathcal{D}_n$. Direct evaluation gives

$$\sum_{n=1}^{32} 2^{|\mathcal{D}_n|} = \sum_{n=1}^{32} 2^{\tau(n)-1} = 539.$$

The supplied census program traverses these masks in lexicographic (n, m) order. For each record it constructs the adjacency relation, applies the proved search to the graph and complement, and serializes either a forbidden cycle or the result of both exhaustive searches. Counting the serialized records gives 415 null-on-both-sides outcomes and 124 witnesses, grouped exactly as stated. Parsing the witness records by n and mask yields Table 1. The program uses only integer gcd operations, bit-set adjacency, and finite branching; no floating-point or probabilistic decision enters the result. $\square$

For example, at $n = 12$ the mask 6 represents $D = \{2, 3\}$, and $(0, 2, 4, 1, 3)$ is an induced 5-cycle in $\text{ICG}_{12}(\{2, 3\})$. Every imperfect record carries such a vertex sequence, on the graph or complement side. Each sequence was normalized under translations, multiplication by units modulo n , cyclic rotation, and reversal; this normalization is only for presentation and does not affect validity.

4 Independent checks and limitations

A separate replay program does not import the census generator. It rebuilds all 539 graphs from the gcd definition, verifies the ordered domain and every adjacency relation, and tests every pair of vertices in all 124 serialized cycles. It accepted every record. As a second implementation check, NetworkX 3.4.2's chordless-cycle enumeration was run on every graph and complement; after scanning 1,602,322 chordless cycles up to the respective decisions, it reported no status mismatch. This agreement is corroboration; the proof of positive cases remains the finite exhaustive search and its completeness lemma.

Elementary recognizers place 329 of the 415 perfect records in at least one of the following classes: bipartite, co-bipartite, complete multipartite, cluster, chordal, co-chordal, or cograph. The

remaining 86 records are only a residue relative to this short list; they are still certified by the complete two-sided search. No divisor-lattice characterization of that residue is claimed.

Most importantly, this note proves nothing for $33 \leq n \leq 64$ or for unbounded n . It does not resolve the general perfectness classification of integral circulant graphs, and it makes no claim that the underlying graph symbols were first enumerated here. Extending the same method to the remaining orders is a concrete next test, while a structural description of the 86-case residue may be more informative than a larger table.

Reproducibility and assistance disclosure

The accompanying archive contains the census source, canonical JSON manifest, complete forbidden-cycle witnesses, a separate replay implementation, and the independent chordless-cycle cross-check. With Python 3, the decisive census is reproduced by

```
python3 evidence/n32_census.py reproduced_manifest.json
python3 evidence/replay_n32.py reproduced_manifest.json replay.json
```

The reported run used CPython 3.13.5; the second cross-check used NetworkX 3.4.2. OpenAI Codex assisted with the computational search, code preparation, verification workflow, and drafting of this note. The mathematical claim is not inferred from model agreement: it rests on the explicit finite algorithm, the completeness proof above, and the serialized exact output.

References

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